File Handling Lab — Step-by-Step Guide

Objective: Learn how to open, read, write, append, and process text files; work with directories and file paths; use **with** statements; and handle common file errors.

Step 1 — Create the Lab Folder

Create a new folder called:

```
file_handling_lab
```

Open this folder in your IDE or code editor.

Step 2 — Create the Python File

Inside the folder, create a Python file called:

```
lab.py
```

Step 3 — Add the Python Code

Paste this complete code into **lab.py**:

```
import os

# PART 1: WRITE, READ AND APPEND

file = open("example.txt", "w")
file.write("Hello, File Handling in Python!\n")
file.write("This is the second line.\n")
file.close()

file = open("example.txt", "r")
print("----- READ FILE -----")
print(file.read())
file.close()

file = open("example.txt", "a")
file.write("This line is appended.\n")
file.close()

file = open("example.txt", "r")
print("----- AFTER APPENDING -----")
print(file.read())
file.close()


# PART 2: LINE-BY-LINE READING

print("----- LINE BY LINE -----")

file = open("example.txt", "r")
for line in file:
    print(line.strip())
file.close()

print("----- READLINES -----")

file = open("example.txt", "r")
lines = file.readlines()
print(lines)
file.close()

# PART 3: FILE AND DIRECTORY OPERATIONS
```

```
print("----- DIRECTORY OPERATIONS -----")

print("Current Directory:", os.getcwd())

if not os.path.exists("data"):
    os.mkdir("data")

with open("data/info.txt", "w") as file:
    file.write("File inside data folder.\n")

print("File exists:", os.path.exists("data/info.txt"))
print("Files in data folder:", os.listdir("data"))


# PART 4: EXCEPTION HANDLING

print("----- EXCEPTION HANDLING -----")

try:
    with open("non_existing.txt", "r") as file:
        print(file.read())
except FileNotFoundError:
    print("File not found!")

try:
    with open("/root/secure.txt", "r") as file:
        print(file.read())
except PermissionError:
    print("Permission denied.")
except FileNotFoundError:
    print("Secure file not found.")
```

Step 4 — Save and Run

Save **lab.py** using Ctrl + S. Open the terminal in the **file_handling_lab** folder and run:

```
python lab.py
```

If you are using VS Code, you can also click the Run (■) button.

Step 5 — Screenshot 1: Write, Read and Append

Check the terminal for the READ FILE and AFTER APPENDING sections. You should see the appended line:

```
This line is appended.
```

Take Screenshot 1 showing **example.txt** in the project explorer and the successful read/append output in the terminal.

Step 6 — Screenshot 2: Line-by-Line Reading

Look for this output:

```
----- LINE BY LINE -----
Hello, File Handling in Python!
This is the second line.
This line is appended.
```

You may also include the READLINES output in the same screenshot. Take Screenshot 2.

Step 7 — Screenshot 3: File and Directory Operations

The program automatically creates a **data** folder and an **info.txt** file inside it. Your project should look like:

```
file_handling_lab/
■■■ lab.py
■■■ example.txt
■■■ data/
    ■■■ info.txt
```

The terminal should show **File exists: True** and **Files in data folder: ['info.txt']**. Take Screenshot 3.

Step 8 — Screenshot 4: Exception Handling

At the bottom of the terminal, look for the missing-file error handling:

```
----- EXCEPTION HANDLING -----
File not found!
```

Depending on your operating system, the second test may display **Permission denied.** or **Secure file not found.**. The important point is that the error is handled without a Python traceback. Take Screenshot 4.

Final Submission — Exactly 4 Screenshots

Screenshot	What to Show
1	Write, Read & Append — example.txt and successful terminal output
2	Line-by-Line Reading and/or readlines() output
3	data folder, info.txt, File exists: True, and folder listing
4	Exception handling output such as File not found! or Permission denied.

Tip: Submit exactly four clear screenshots. Avoid taking one large screenshot of the entire program.